

Environment & Society — Week 03 Practice

Questions

1. [L5] Define **development** as it's used in this course, and name the three broad categories used to describe a country's development level.
 2. [L5] What does the "Level 1–4" income scale (breakpoints at \$2/\$8/\$32 per day) reveal about how the world's population is actually distributed? Why does the professor call the simple "haves and have-nots" framing misleading?
 3. [L5] Why do geographers consider binary terms like "First World / Third World" or "developed / developing" to be problematic descriptions of the real world?
 4. [L5] Define **colonialism**.
 5. [L5] A country has just begun intensively exporting a single major natural resource and is undergoing rapid political change as a result. Using Rostow's five-stage model, which stage best fits this description, and why?
 6. [L5] Name the three theories of development covered in this week's material, along with the person credited with each.
 7. [L5] According to World Systems Theory, what distinguishes **core**, **semi-periphery**, and **periphery** countries, and how does Wallerstein describe the relationship between them?
 8. [L5] A critic points out that the United States and Canada were both British colonies, yet both became wealthy — unlike many other former colonies. Which theory does this observation challenge, and why?
 9. [L6] Distinguish between **GDP** and **GNI**.
 10. [L6] Explain the three main problems with using GNI per capita as a measure of development.
 11. [L6] Ecuador's average income is \$4,510 and milk costs \$1.02/L there; Canada's average income is \$41,950 and milk costs \$2.25/L. Using these figures, explain why a Purchasing-Power-Parity-based comparison puts Canada at roughly 4× richer than Ecuador, rather than the roughly 10× suggested by raw income alone.
 12. [L6] Define **Purchasing Power Parity (PPP)**, and describe its general effect on the apparent income gap between rich and poor countries.
 13. [L6] What three dimensions does the **Human Development Index (HDI)** combine into a single score?
 14. [L6] Norway's 2019/2020 HDI is 0.957 and Niger's is 0.394. Express each country's "short-fall" from the theoretical maximum of 1.000 as a percentage.
 15. [L6] Name two things the HDI still fails to capture well, and identify a related index that attempts to address one of them.
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Answers

1. **Answer:** Development is *a process that brings about changes in economic prosperity and quality of life; an improvement in the material conditions of life*, traditionally measured by economic criteria but increasingly measured more holistically (health, education, etc.). The three broad categories are the **more developed world**, **less developed world**, and **least developed world**. — [Notes, Part 1 “What is development?”; L5 S5, S8][T p.174–175]
2. **Answer:** On the Level 1–4 scale, roughly 1 billion people sit at Level 1 (poorest, <\$2/day) and roughly 1 billion at Level 4 (richest, >\$32/day) — but the remaining ~5–6 billion of the world’s ~8 billion people fall in Levels 2–3, in between. The “haves and have-nots” framing is misleading because it implies only two groups exist, when in fact most of humanity occupies a middle ground between the extremes. — [Notes, Part 1 “What is development?”; L5 S4, S7]
3. **Answer:** These binary terms imply the world splits cleanly into two opposing groups, but the Level 1–4 data shows most people and countries fall somewhere in between the extremes — a continuum, not a binary. The “Global North/South” line also produces geographic absurdities (e.g. Australia and New Zealand, in the southern hemisphere, are grouped with the “North” because the line tracks development level, not literal hemisphere). — [Notes, Part 1 “Differentiating levels of development”; L5 S7–S8]
4. **Answer:** Colonialism is *the forceful appropriation of foreign territory; usually established and maintained through military and political structures; creates unequal cultural and economic relations; usually involves the displacement of indigenous populations*. — [Notes, Part 1 “Differentiating levels of development”; L5 S9][T p.184]
5. **Answer: Stage 3, Take-off (to sustained growth)** — this stage is defined by exploitation of major resources and possibly rapid political change, matching a country that has just begun intensively exporting one major resource amid political upheaval. (Note: exploiting a single resource for export can also *start* in Stage 2, Preconditions for take-off, if it’s a smaller/localized development — the key distinguishing feature of Stage 3 is that the resource exploitation is now driving *sustained* growth.) — [Notes, Part 1 “Theories of development”; L5 S13][T p.180, Box 5.2]
6. **Answer: Modernization Theory** (Walter Rostow, economist), **Dependency Theory** (many contributors/variants, no single named originator), and **World Systems Theory** (Immanuel Wallerstein, sociologist). — [Notes, Part 1 “Theories of development”; L5 S11][T p.185]
7. **Answer: Core** countries are the more developed places (e.g. Europe, North America, Japan, Australia); **semi-periphery** countries are less developed (e.g. China, India, Brazil, Mexico); **periphery** countries are least developed (e.g. much of sub-Saharan Africa). Wallerstein describes them as linked together in “an interdependent system of countries... by an economic and political competition that shapes relations” between the three zones — historically, core countries exploited peripheral/semi-peripheral ones for labour and resources, and this dependency-like dynamic persists today through globalization. — [Notes, Part 1 “Theories of development”; L5 S16, S18][T p.185–186]
8. **Answer:** This challenges **Dependency Theory**, since it predicts that former colonies remain economically dependent on and dominated by their former colonizer. The US/Canada

case is a clear counterexample — both were British colonies but became wealthy and independent rather than staying dependent. The lecture raises this as an open, unresolved critique (with a speculative, unconfirmed suggestion that racial dynamics might be part of the explanation). — [Notes, Part 1 “Theories of development”, and Gaps & conflicts; L5 S15]

9. **Answer:** **GDP** is a monetary measure of the market value of all goods and services produced *within* a country over a given period. **GNI** is the same, but adds income generated by that country’s investments *abroad* (e.g. a Canadian-owned factory in China). Both are divided by population to get a per-capita proxy for average income. — [Notes, Part 2 “Measuring development: economic tools”; L6 S4][T p.178]
10. **Answer:** (1) GNI per capita only captures the **formal economy** — it misses informal/untracked activity, which can be as much as ~80% of economic activity in some less-developed countries, causing GNI to *underrepresent* their true economic size. (2) It is an average, so it hides **inequality within a country** (e.g. billionaires vs. people in poverty in the same nation). (3) It doesn’t account for **differences in cost of living** between countries. — [Notes, Part 2 “Measuring development: economic tools”; L6 S6]
11. **Answer:** By raw income, Canada looks about $9.3\times$ richer than Ecuador ($\$41,950 \div \$4,510$). But converting each income into how much milk it can actually buy locally gives Ecuador $\approx 4,421$ L ($\$4,510 \div \1.02) and Canada $\approx 18,644$ L ($\$41,950 \div \2.25) — a ratio of about $4.2\times$ ($18,644 \div 4,421$). Because Canada’s cost of living (milk, in this example) is also higher than Ecuador’s, part of Canada’s raw income advantage is “absorbed” by higher local prices, so the *real* purchasing-power gap is smaller than the raw income gap suggests. — [Notes, Part 2 “Purchasing Power Parity (PPP)”; L6 S7]
12. **Answer:** PPP is *a tool that measures the relative cost of a common market basket of goods and services for comparing cost of living between countries*, used alongside aggregate measures like GDP/GNI per capita. Its general effect is to **compress** the apparent income gap between richer and poorer countries — narrowing but not eliminating it (the underlying core-periphery pattern of uneven development still holds after adjustment). — [Notes, Part 2 “Purchasing Power Parity (PPP)”; L6 S6, S8][T p.179]
13. **Answer:** **Income** (GNI per capita), **health** (life expectancy), and **education** (years of schooling). — [Notes, Part 2 “Combining economic & social measures: the HDI”; L6 S13][T p.180 sidebar, 183]
14. **Answer:** Norway: $1.000 - 0.957 = 0.043 \rightarrow 4.3\%$ shortfall. Niger: $1.000 - 0.394 = 0.606 \rightarrow 60.6\%$ shortfall. — [Notes, Part 2 “Combining economic & social measures: the HDI”; L6 S14][T p.181, 183, Table 5.1] (*Not machine-verified — per this course’s rules, no computation check is required; this is simple arithmetic directly matching the textbook’s own stated shortfall figures.*)
15. **Answer:** HDI does not account for **gender inequality** (e.g. unpaid domestic labour, or mothers more often taking time off for childcare) — addressed, imperfectly, by the related **Gender Development Index (GDI) / Gender Inequality Index (GII)**. It also fails to capture harder-to-quantify factors like **political freedom** (freedom of expression/protest) or environmental quality and safety — these remain largely unmeasured by any current index.

— [Notes, Part 2 “Addressing what HDI still misses: IHDI & GII”; L6 S15–16]